Evaluation for Generality of Natural Japanese Sentence Generation Method Using Inductive Learning

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Abstract

A lot of unnatural sentences exist in the Japanese sentences which the computer generates in the machine translation, the dialogue processing and so on. The user has dissatisfaction for it. From the point of view of this situation, we previously proposed the method which paraphrases an unnatural Japanese sentence generated by a computer to its natural Japanese sentence with inductive learning. In order to confirm the effectiveness of our proposed method, we constructed the experimental system and carried out the evaluation experiment. The rate of the number of natural sentences in the number of all input sentences improved from 32.4% to 55.3%. From the result of this evaluation experiment, we have confirmed the effectiveness of our method. In this paper, first of all, we explain the outline of our proposal method, and consider the effectiveness of our method through the experimental results.

Keywords

inductive learning, natural language processing, machine translation, naturalness, paraphrasing

1. Introduction

Recently, we often treat a sentence with a computer, because a machine translation system and a dialogue processing system have spread. However, a lot of unnatural sentences exist in the sentences that the computer generates in the machine translation, the dialogue processing and so on. This is a problem which does not depend on the specific language. The user has dissatisfaction for it. From the point of view of this situation, we propose the method that paraphrases an unnatural Japanese sentence generated by a computer to its natural Japanese sentence using inductive learning [1].

The paraphrase has been researched so far. However, the traditional methods [2,3,4] are the methods by which the sentence is paraphrased for a certain purpose and the methods [5] to select the term used in a translation. Most of the traditional methods paraphrase from a correct sentence and word to other expressions of them. Furthermore, though the research [6] on the naturalness of other language exists, the research on the naturalness of a Japanese sentence hardly exists. Our proposed method is different from the traditional method, because it paraphrases an unnatural Japanese sentence to a natural Japanese sentence.

Our proposed method uses inductive learning. We consider that our approach can resolve the problems of the Rule-based approach. The Rule-based approach is not robust. The system based on this approach can deal with a limited situation that a developer expects. However there are many situations which the developer can not expect in the world which natural language can express. Therefore this approach can not deal with many varieties of language phenomena. Our proposed method can adapt to a target by adapting a rule dynamically using inductive learning, even if the target changes. Moreover it does not need a huge corpus because it can acquire a needed rule from a comparative few examples using inductive learning.

In this paper, we explain the outline of our proposed method, and consider the effectiveness of our method through the experimental results.

2. Outline of This Method

Figure 1 shows an outline of this method. First, the user inputs a sentence. The input sentence is a Japanese sentence which the computer generated. Second, the system carries out the morphological analysis. Most of input sentences are unnatural Japanese sentences, and they contain the mistake a lot. Even if they carry out the syntactic analysis and the semantic analysis, a correct analytical result is not often obtained. Therefore, the system carries out only the morphological analysis. Third, in the rule search process, the system examines whether there are rules that match to the input sentence in the rule dictionary. There are two kinds of rules in the rule dictionary. The first are called input rules. And the second are called paraphrase rules. The input rule is a rule with which the input sentence is replaced by the variable as much as possible. The paraphrase rule is a rule with which the correct sentence is replaced by the variable as much as possible. And an input rule and a paraphrase rule are pairs. Each rule is extracted in the learning process. Fourth, in the rule selection

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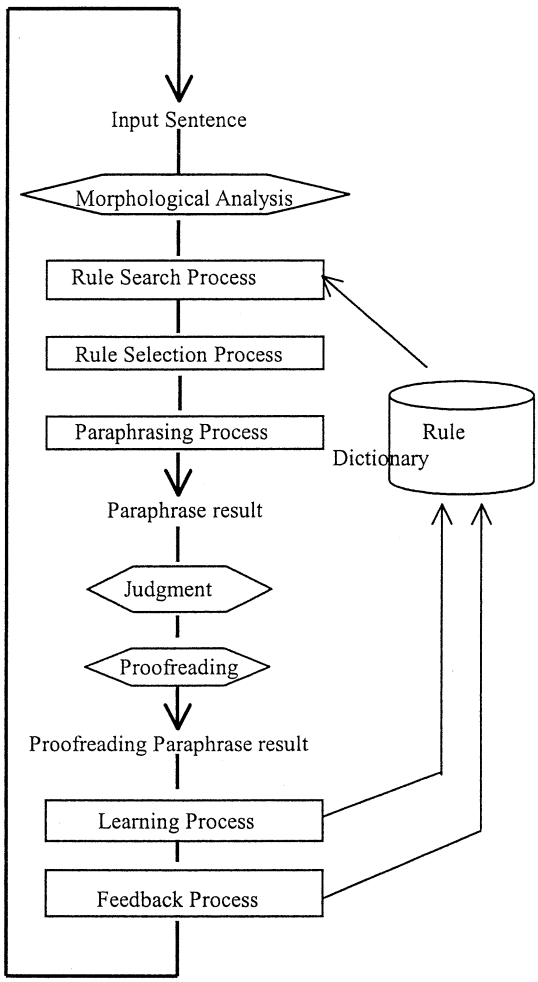


Figure1: Outline of our method

process, the system selects best rules from among rules matched in the rule search process. Fifth, in the paraphrase process, the system paraphrases the input sentence using paraphrase rules. Sixth, in the proofreading process, the user proofreads the paraphrased sentences. Seventh, in the learning process, the system extracts various rules from the paraphrase examples using inductive learning. At last, in the feedback process, the system increases the promising degrees of correctly paraphrased rules and decreases the promising degrees of erroneously paraphrased rules.

3. Processing of This Method

3.1. Learning Process

In this process, the system extracts various rules from the paraphrase examples by inductive learning. The acquisition way of the rule includes three ways.

1. The system compares the input sentence with the proofreading sentence in each morpheme. If the same morpheme exists in two sentences, the morpheme is replaced with the variable.

2. From the head of the sentence sequentially, the system compares the input sentences with the proofreading sentence in each morpheme. When a different morpheme appears in two sentences, a morpheme string from the previous morpheme to the end of sentence is acquired.

3. From the end of the sentence sequentially, the system compares the input sentences with the proofreading sentence in each morpheme. When the independence word which appears first in each sentence is the same, the morpheme string from the independence word to the end of sentence is acquired. The morpheme string in the first half is acquired at the same time.

The rule acquired by the step 1 is called a sentence rule (S-rule). The rule acquired by the step 2 and the step 3 is called a word rule (W-rule). Figure 2 shows an example of learning process. In Figure 2, a capital letter means the independence word.

3.2. Rule Search Process

In this process, the system examines whether there are rules that match to the input sentence in the rule dictionary. The input sentence and the input rule are performed the morphological analysis and they are compared in each morpheme. The input rule might contain the variable. In that case, when the part of speech is corresponding, the system determines the same morpheme. Only when all morphemes are corresponding through the sentence, the system determines that the input rule matched to the input sentence. This process is carried out to all input rules maintained in the rule dictionary at that time.

3.3. Rule Selection Process

In this process, the system selects best rules from among rules matched in the rule search process. This process has two stages. The first stage is a score calculation of variable parts. The second stage is a promising degree calculation of input rule.

3.3.1. Score Calculation of Variable Parts

When both S-rule and W-rule are matched in the rule search process, S-rule is applied. Moreover, when both the rule which does not contain the variable and the rule which contains the variable are matched in the rule search process, the former is applied.

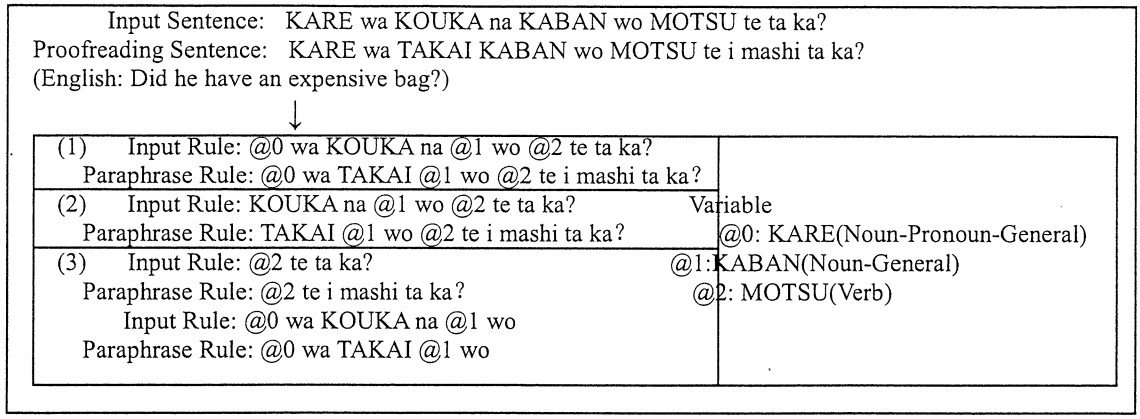


Figure2: Example of Learning Process

Therefore, rules are classified into four ranks. The rule ranks are S-nv-rule(not contain the variable), S-v-rule(contain the variable), W-nv-rule(not contain the variable) and W-v-rule(contain the variable). The order of higher certainty degree is S-nv-rule, S-v-rule, W-nv-rule and W-v-rule. S-rule which can be applied to one input sentence is only one rule. W-rule which can be applied to one input sentence is one or more rules.

Next, we explain the case which plural rules of the same rank are matched in the rule search process. Variable parts of each rule have morphemes which have been correctly applied before. Part of speech, inflection, and concept number are also maintained about these morphemes. The Concept Number is referred from "Bunruigoihyou"[7]. To calculate the score of the variable part, the morpheme of the input sentence which corresponds to the variable part is paid attention. Table 1 shows how to calculate the score of the variable part.

Table1: How to calculate the score

Condition	Score
The morpheme has been correctly applied to this variable part before.	+2
The morpheme has the same concept as the morpheme which has been correctly applied before.	+1
Others	+0

According to the score shown in Table 1, the total point of each input rule is calculated. The value that this total point was divided by "Number of variables× 2" is assumed the score of each input rule[1].

[1] "Number of variables×2" is full marks of the total point.

Therefore, a final score of each input rule is calculated through the following equations:

$$\text{Final Score} = \frac{\text{Total point}}{2\times(\text{Number of variable})} \quad (1)$$

This final score is calculated about each input rule, and the scores are compared. If the rule which has the most score is only one, the rule is applied. And the rule selection process finishes. If the rule which has the most score is not only one, the system moves to the second stage.

3.3.2. Promising degree calculation of input rule

The promising degree is calculated by the promising evaluation function defined as:

$$PEF = \alpha \square CF - \beta \square EF \quad (2)$$

,where α and β are coefficients, the PEF is the promising evaluation function, CF is the frequency of the correct paraphrase and EF is the frequency of the erroneous paraphrase.

3.4. Paraphrasing Process

When the input rule which matches to the input sentence exists, the input sentence is paraphrased according to the paraphrase rule corresponding to the input rule. If the input rule which matches to the input sentence does not exist, the input sentence is remained.

3.5. Feedback Process

In this process, the system increases the promising degree of correctly paraphrased rules and decreases

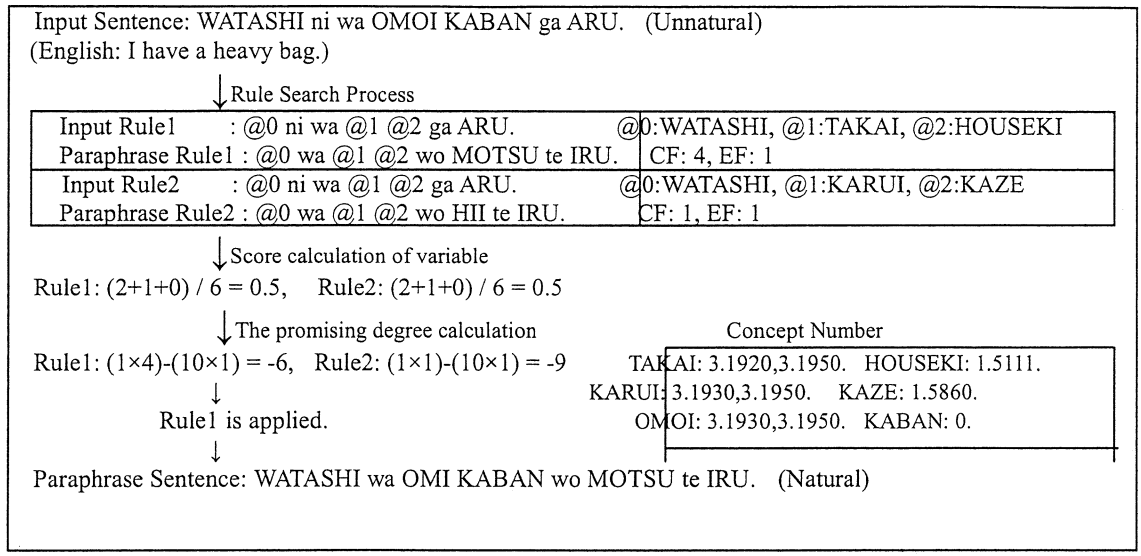


Figure3: Example of the process of our method

the promising degree of erroneously paraphrased rules. The promising degree is calculated by PEF of the equation (2). In this process, CF and EF are updated. The correct paraphrase is defined as one in which the paraphrased result and the proofread result are the same. The erroneous paraphrase is defined as one in which the paraphrased result and the proofread result are different. The system adds one to the value of CF when the rule is used for the correct paraphrase. The system adds one to the value of EF when the rule is used for the erroneous paraphrase.

Finally, Figure 3 shows the example of the process of our method. The rules which match to the input sentence are assumed to be two (Rule 1 and Rule 2). Because both rules are the same rank, the rule applied to this input sentence is selected by the score calculation and the promising degree calculation. The system selected rule 1 in the rule selection process. As a result, an unnatural input sentence was paraphrased to a natural sentence.

4. Experiment

4.1. Standards for Evaluation

We evaluate the performance of our method using translation examples of the English travel conversations "at the check-in", "on the plane", "at the airport" and "on the telephone" from English and Japanese. We call "at the hotel", "on the plane", "at the airport" and "on the telephone" Scene 1, Scene 2, Scene 3 and Scene 4 respectively. In these experiments, a correct paraphrase result defines an exact same character string or a very similar expression as that of a translator. These evaluation standards are very high. However translations like this are required by today's users.

4.2. Data and Procedure

In these experiments, we used 456 translation examples of Scene1, 345 translation examples of Scene2, 606 translation examples of Scene3 and 296 translation examples of Scene4. These data were taken from 10 English travel books for Japanese. These books are [8,9,10,11,12,13,14,15,16,17].

And we used two other machine translation systems which are developed by A and B companies for confirming generality of our method. We call these two machine translation systems A,B respectively. Chasen[18] was used for the morphological analysis. Chasen is a famous Japanese morphological analysis tool.

In order to confirm whether inductive learning is effective in various data, the rule dictionary was empty at first. The values of the coefficients in equation (2) are $\alpha=1$ and $\beta=10$. These values are decided by another experiment [19].

4.3. Results and Considerations

Figure 4 and Figure 5 show the rate of the number of natural sentences in the number of all input sentences in A and B respectively. We call the rate of the number of natural sentences in the number of all input sentences the natural rate. Before experimented, the total natural rates of A and B are 32.4% and 29.7% respectively. And after experimented, the total natural rates of A and B are 48.1% and 46.3%

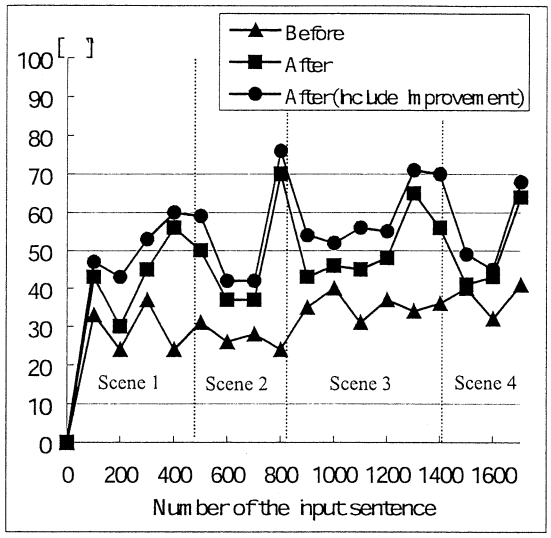


Figure4: The rate of the number of natural sentences in the number of all input sentences in A

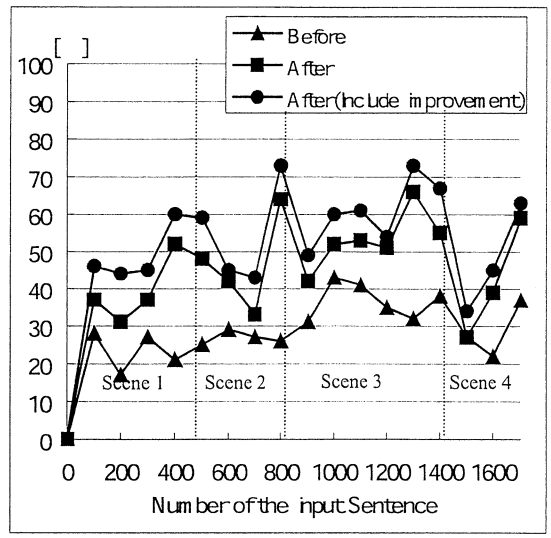


Figure5: The rate of the number of natural sentences in the number of all input sentences in B

respectively. There are sentences to which a part is improved in the paraphrase results, though it is not natural through the sentence. When those sentences are included in natural sentences, the total natural rates of A and B are 55.3% and 54.1% respectively. Both the total natural rates improve by more than 20 points. From this result, we have confirmed generality of our method. Table 2 shows the total natural rates of A and B.

Table2: Total natural rates of A and B

	Before	After	After (include improvement)
A	32.4%	48.1%	55.3%
B	29.7%	46.3%	54.1%

The more the input sentence reaches in the latter half of each scene, the more the natural rate increases. For instance, in A, the natural rate in 100 sentences from 500th to 600th is 37.0%. However the natural rate in 100 sentences from 700th to 800th is 70.0%. Therefore, we confirm that the learning capability of our method is high enough so that this system can adapt to Scene 1, Scene 2, Scene 3 and Scene 4.

The rate of the side effect of this method for A and B are only 2.47% and 2.23%. The side effect means that the system paraphrases a natural sentence to an unnatural sentence. This means that the system does not have an adverse effect on a natural sentence. This is the maximum advantage of this system. Table 3 and Table 4 show examples of the correct paraphrase result and the erroneous paraphrase result respectively. In Table 3, all input sentences are

Table3: Examples of the correct paraphrase result

1	Input Sentence Paraphrase Sentence English	まさしく砂糖をお願いします。 砂糖だけをお願いします。 Sugar, please.
2	Input Sentence Paraphrase Sentence English	私には、700ドルがある。 700ドル持っています。 I have 700 dollars.
3	Input Sentence Paraphrase Sentence English	「領域コードは」06です。 「市外局番は」06です。 The area code is 06.

Table4: Examples of the erroneous paraphrase result

1	Input Sentence Paraphrase Sentence English	オペレータをお願いします。 オペレータを下さい。 Operator, please.
2	Input Sentence Paraphrase Sentence English	テレビを持っているか? テレビは空いていますか? Do you have a TV?
3	Input Sentence Paraphrase Sentence English	旅行の目的は何であるか? 旅行の目的はいくらですか? What is purpose of your visit?

unnatural. And all paraphrase sentences are natural. In Table 3, the rules used for 1 and 2 are S-v-rule. The rule used for 3 is W-nv-rule. In Table 4, input

sentences in 1and 3 are natural and input sentence in 2 is unnatural. But all paraphrase sentences are unnatural. The cause is mis-application of rules.

5. Conclusions

In this paper, we proposed the method which paraphrases an unnatural Japanese sentence generated by a computer to its natural Japanese sentence with inductive learning. We consider that our approach can resolve the problems of the Rule-based approach. In order to confirm the effectiveness of our method, we carried out the evaluation experiment. The rate of the number of natural sentences in the number of all input sentences in A and B improved from 32.4% to 55.3% and from 29.7% to 54.1% respectively. Both the total natural rates improve by more than 20 points. The rate of the side effect of this method for A and B are only 2.47% and 2.23%. From the result of this evaluation experiment, we have confirmed the effectiveness of our method. Our method was adequate enough to obtain highly-correct and highly-qualitative paraphrase results in four scenes. And we consider that our method can also be adequate enough to obtain similar performance for various scenes. Because, in these experiments, the initial rule dictionaries were empty and it could only learn the context from paraphrase examples through its own learning capability. We have also confirmed generality of our method. In these experiments, we used two other machine translation systems. As for both systems, the natural rates increase. In the future, we plan to experiment using various data of many other fields.

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